EE 3001 Telecommunication System Design with DSP Capsule

Capsule Project

Objective

In this project, you will design a communication system using Universal Software Radio Peripheral (USRP), as shown in Figure 1. The project includes both telecommunication and DSP components. The telecommunication technique is Binary Phase Shift Keying (BPSK), and the DSP technique is based on Linear Predictive Coding (LPC) represented by Line Spectral Frequencies (LSFs).

The goal of the project is to design an end-to-end wireless speech communication system. The communication system should use a speech signal. Instead of transmitting the full speech waveform directly, the speech signal will be processed, represented using a compact set of parameters, transmitted in binary form, and approximately reconstructed at the receiver side.

Project Description

In many practical communication systems, transmitting raw speech samples directly is not efficient. Speech signals can be represented by a smaller number of meaningful parameters while still preserving the main characteristics of the signal. In this project, a speech signal sampled at 8 kHz will be used as the input to the speech representation stage of the system.

You will be given a corrupted speech signal sampled at 48 kHz. This signal contains unwanted high- frequency components. Therefore, the signal must be processed appropriately so that a speech signal sampled at 8 kHz is obtained for the LPC/LSF speech-processing stage. The later BPSK transmission stage may use a much higher waveform sampling rate for pulse shaping, modulation, and USRP operation.

After sampling-rate reduction, the speech signal will be analyzed frame by frame using LPC. The LPC model will be represented in terms of LSF parameters, which are more suitable for quantization and transmission. Additional frame parameters such as gain, pitch, and voiced/unvoiced information may also be included in the transmitted representation.

The resulting binary data will be transmitted using BPSK over a wireless communication link. At the receiver, the signal will be demodulated, the transmitted parameters will be recovered, and the speech signal will be approximately reconstructed. The overall performance of the system will depend on the quality of signal preparation, parameter extraction, transmission, reception, and reconstruction.

Figure 1 shows the overall block diagram of a communication system. You are supposed to implement a wireless communication system using USRP. You need to investigate the effect of noise in the communication system, and evaluate the quality of the reconstructed speech under different conditions. The transmitted data is exposed

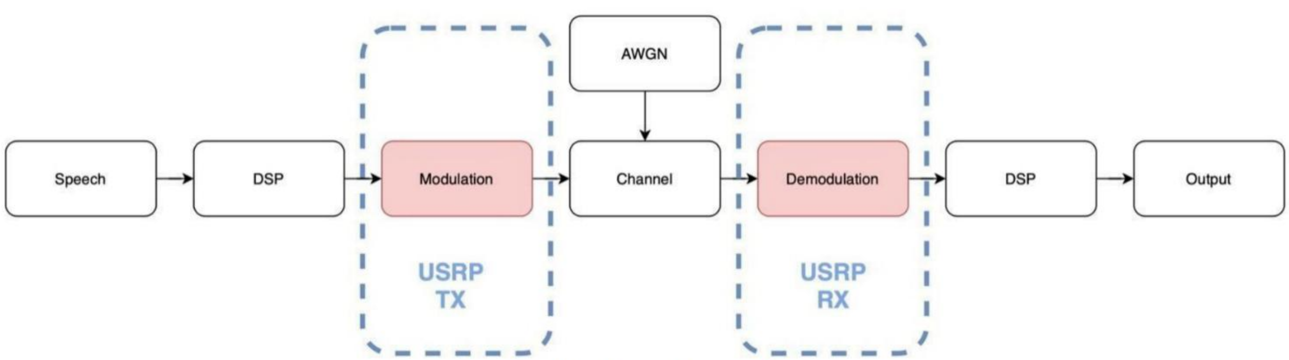

Figure 1. Pipeline of project

to noise during transmission, and the final evaluation will be based on the correctness and performance of the complete system. Although it is not a requirement, error correction coding (ECC) may also be applied as an extra improvement (implementation of the ECC will be extra credit).

Expected System

1. Corrupted speech input sampled at 48 kHz
2. Preparation of a speech signal sampled at 8 kHz
3. DSP-based speech parameter extraction
4. Binary representation of the selected parameters
5. BPSK modulation and wireless transmission
6. BPSK demodulation and bit recovery
7. Reconstruction of speech parameters
8. Approximate speech synthesis at the receiver

Project Report

1. Problem definition and motivation
2. Description of the signal preparation stage
3. Telecommunication technique used in the system
4. DSP technique used in the system
5. Description of the transmitter and receiver structure
6. Data size comparison before and after DSP-based representation
7. Performance evaluation of the communication system
8. Quality of reconstructed speech and discussion of results

Rules

- Project groups can have a maximum of two people.
- You have to use USRP for the modulation and demodulation part.
- You have to evaluate the effect of channel conditions on the transmission and receiving process in your experiments.
- At the end of the semester, you have to do a demo.
- The implementation should reflect both the telecommunication and DSP aspects of the project.

Schedule:

Progress of DSP part	Due date 08/05/2026
Progress of Telecom part	Due date 11/05/2026
Demo & Project Report	Due date 01/06/2026

Evaluation rubric:

DSP Progress	2.5
Telecom Progress	2.5
Project Report	5
Project Demo	9